Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		This is the energy needed to eject an electron (1) from the surface **or** the metal **or** rubidium (**not** the atom) (1)	2			2		
	(b)		Any electron ejected is ejected by a single photon or equivalent (1) Photon energy [or hf] must be [at least] equal to ϕ (1) So $hf > \phi$ [for emission] (1)	1 1	1		3		
	(c)	(i)	Mean = 0.377 [V] Accept 0.376 [V] **or** 0.3765 [V] **or** 0.38 (V) (1) ± 0.011 [V] **or** 0.0110 [V] **or** 0.01 [V] (1) Answer + uncertainty expressed to an appropriate number of sig figs i.e. 0.377 (or 0.376) V ± 0.011 V or 0.38 ± 0.01 [V] (1) Must use earlier calculations		3			3	3
		(ii)	$E_{\text{k max}}$ calculated for **one** stopping pd i.e. 0.366 V → 5.86×10^{-20} J, **or** 0.388 V → 6.21×10^{-20} J **or** (0.377 V → 6.03×10^{-20} J) (1) Einstein's equation used correctly, either with hf and **one** $E_{\text{k max}}$ value including mean to give ϕ, **or** with hf and $\phi = 3.62 \times 10^{-19}$ J to give $E_{\text{k max}}$ e.g. $\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 5.86 \times 10^{-20}$ (1) $= 3.62 \times 10^{-19}$ [J] (1) **or** $\phi = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 6.21 \times 10^{-20}$ (1) $= 3.58 \times 10^{-19}$ [J] (1) **or** $E_{\text{k max}} = 6.63 \times 10^{-34} \times 6.34 \times 10^{14} - 3.62 \times 10^{-19}$ (1) $= 5.83 \times 10^{-20}$ [J] (1)						